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Thinly coated hollow core fiber for improved thermal phase-stability performance

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The optical phase accumulated when light propagates through an optical fiber changes with temperature. It has been shown by various authors that this thermal phase sensitivity is significantly smaller in hollow core fibers (HCFs) than in standard single-mode fibers (SSMFs). However, there have been considerable differences in the level of sensitivity reduction claimed, with factors in the range $\times 3$ to $\times 20$ improvement for HCFs relative to SSMFs reported. Here we show experimentally that this large variation is likely attributable to the influence of fiber coating, which is exacerbated in HCFs with a relatively thin silica glass outer wall (e.g., the wall thickness is typically just 20 μm in a 125 μm diameter HCF). Further, we show that the coating also causes the optical phase stability to suffer from relaxation effects, which have not been previously discussed in the HCF literature, to the best of our knowledge. In addition to demonstrating these relaxation effects experimentally, we analyze them through numerical simulations. Our results strongly suggest that they originate from the viscoelastic properties of the coating. To minimize the adverse effects of the coating, we have fabricated a HCF with a relatively thick wall ($\sim 50 \mu\text{m}$) and a very thin coating (10 μm). This results in an almost 30-fold reduction in HCF thermal phase sensitivity relative to SSMFs – a significantly lower sensitivity than in previous reports. Moreover, our thinly coated HCF exhibits no discernable relaxation effects while maintaining good mechanical properties. © 2021 Optical Society of America

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The optical phase accumulated during the propagation of light through an optical fiber varies due to environmental changes (temperature, vibrations, etc.), which is undesirable for many applications. For example, in interferometers, which lie at the heart of some of the most precise instruments that exist today (e.g., for use in metrology, fiber gyroscopes [1], and gravitational wave detectors [2]), this unwanted thermal effect limits the achievable measurement sensitivity and accuracy.

The temperature sensitivity of the accumulated phase of light propagating through a length L of fiber ($\varphi = 2\pi n_{\text{eff}} L / \lambda$) can be

quantified through

$$S_\varphi = \frac{1}{L} \frac{d\varphi}{dT} = \frac{2\pi}{L\lambda} \frac{d(n_{\text{eff}}L)}{dT} = \frac{2\pi}{\lambda} \left(\frac{dn_{\text{eff}}}{dT} + \frac{n_{\text{eff}}}{L} \frac{dL}{dT} \right), \quad (1)$$

where n_{eff} is the effective refractive index, λ is the wavelength in vacuum, and T is the temperature. Please note that we normalize S_φ to unit physical fiber length while, in the literature, it is sometimes normalized to unit optical path length (which in silica fibers is about 1.45 times larger while, in hollow core fibers (HCFs), it is almost identical to the physical length L).

In Eq. (1), the first term on the right-hand side represents contributions from changes due to the thermo-optic effect, while the second term relates to the thermally induced fiber elongation. In solid glass-core silica-based standard single-mode fibers (SSMFs), the first term accounts for 95% of the contribution to S_φ , which is typically around 48 rad/m/K at a wavelength of 1550 nm [3,4]. In HCFs, this term is negligible, leading to the thermal sensitivity of the HCF being significantly lower than that of a SSMF and typically limited by the second term in Eq. (1) [thermally induced fiber elongation]. Considering the widely used value of the coefficient of thermal expansion (α_{silica}) of fused silica glass at a room temperature of 0.55 ppm/K [5], the second term in Eq. (1) should in a HCF ($n_{\text{eff}} \sim 1$) induce $S_\varphi = 2.2$ rad/m/K. This is close to the values published for HCFs [6] of 0.52 ppm/K (corresponding to a S_φ of 2.1 rad/m/K). The coefficient α_{silica} gets smaller at lower temperatures and crosses zero around -80°C [5], enabling $S_\varphi = 0$ (zero phase sensitivity) of the HCF to be achieved. This has been confirmed experimentally, with $S_\varphi = 0$ being reported at -71°C [7].

The above analysis strictly speaking applies for uncoated fibers only. The acrylate coatings typically used on optical fibers for protection and to improve their properties have significantly higher coefficients of thermal expansion than those of fused silica. When a fiber is heated, the coating elongates the fiber beyond what is expected for fused silica. Dangui *et al.* [8] used a simple model to account for this coating contribution. The thermal coefficient of expansion of a single-material coated fiber in this model is

$$\alpha_{\text{coated_fib}} = \frac{\alpha_{\text{silica}} E_{\text{silica}} A_{\text{silica}} + \alpha_{\text{coating}} E_{\text{coating}} A_{\text{coating}}}{E_{\text{silica}} A_{\text{silica}} + E_{\text{coating}} A_{\text{coating}}}, \quad (2)$$

where A denotes the cross-sectional area and E denotes the Young's modulus of silica glass and the coating material. Indeed, this model can be straightforwardly extended to more layers [8]. We will consider room temperature (25°C) and the representative coating parameters given in Ref. [3]: $E_{\text{coating}} = 158$ MPa, $\alpha_{\text{coating}} = 180$ ppm/K, $E_{\text{silica}} = 72.5$ GPa, and $\alpha_{\text{silica}} = 0.55$ ppm/K. Given $E_{\text{coating}} \ll E_{\text{silica}}$ and considering $A_{\text{coating}} \sim A_{\text{silica}}$, we can neglect the second term in the Eq. (2) denominator, simplifying Eq. (2) to

$$\alpha_{\text{coated_fib}} \cong \alpha_{\text{silica}} + \alpha_{\text{coating}} \frac{E_{\text{coating}} A_{\text{coating}}}{E_{\text{silica}} A_{\text{silica}}}. \quad (3)$$

Considering the above numbers, we get

$$\alpha_{\text{coated_fib}} \cong 0.55 \text{ ppm K}^{-1} + 0.39 \text{ ppm K}^{-1} \frac{A_{\text{coating}}}{A_{\text{silica}}}. \quad (4)$$

We see that for a coating A_{coating} similar or larger than A_{silica} , the coating may strongly increase the overall fiber expansion. We believe this coating contribution lies behind most of the discrepancies in the S_ϕ measured for HCFs in the literature (e.g., [3,8,9]). Unfortunately, we cannot quantify this, as the data in the literature do not give all the necessary information, e.g., the coating diameter is often not given. However, it seems that the relatively higher values of thermal sensitivity are measured with fibers with relatively small glass fiber cross-sectional areas [8,9]. By contrast, the smaller S_ϕ values reported [3,4,6,10] are mostly on HCFs manufactured at the University of Southampton, where relatively large glass cladding diameters (e.g., 180–260 μm) are generally used with a corresponding glass cross-sectional area that is about two times larger than for other HCFs, e.g., as measured in [8]. Additionally, the discrepancy in the reported values is often further exacerbated by different normalization approaches as we mentioned earlier (giving a further discrepancy of about 1.45 times).

In this Letter, we first manufactured a HCF with a very thin coating (10 μm) and a relatively thick glass wall (50 μm), which reduces the coating's contribution to $\alpha_{\text{coated_fib}}$ to less than 18%, providing both a practical solution (as the fiber is not as fragile as it be for an uncoated fiber [3]) and a low thermal sensitivity. We measured the thermal sensitivity and compared it to that of the standard-thickness-coated HCF reported earlier [6]. To further study the coating contribution to S_ϕ , we also measured a HCF that would be expected to have a significantly larger coating contribution than in our previously studied HCF. This is achieved by using a coating typically employed as a secondary fiber coating in dual-coated fibers that have a significantly larger Young's modulus. We show that, besides degrading S_ϕ , such a coating also introduces time-dependent creep, which we confirm by comparing measured data with simulations.

The manufactured thinly coated HCF was of a NANF geometry [11] with a glass silica jacket tube of 185/70 μm outer/inner diameter; see Fig. 1. The coating was 10 μm thick. Equation (4) then gives $\alpha_{\text{coated_fib}} \cong 0.65$ ppm/K, which is only marginally larger than that expected of the uncoated fiber (0.55 ppm/K).

The second sample we measured was chosen to get an appreciable contribution from the coating. As none of the fibers available to us had a small A_{silica} , we used a fiber with a relatively large A_{silica} , but dual-coated, as the secondary fiber

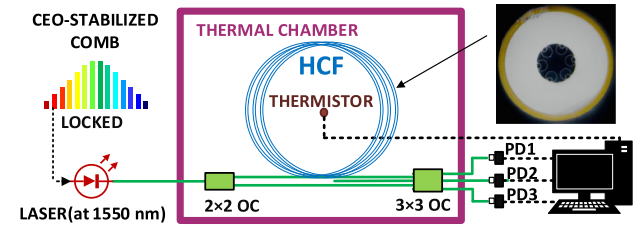


Fig. 1. Measurement setup used to characterize the fiber thermal sensitivity and photograph of the manufactured thinly coated HCF. OC, optical coupler; PD, photodetector.

coating typically has a significantly larger Young's modulus than the single coating, creating a more prominent effect, as follows from Eq. (3). Neglecting the influence of the soft inner coating and using $E_{\text{out_coating}} = 780$ MPa, Eq. (4) gives $\alpha_{\text{dual_coated_fib}} \cong 0.93$ ppm/K, which shows a significant contribution from the coating to the overall fiber elongation ($\sim 70\%$ larger than that of the silica glass).

Figure 1 shows the experimental setup used to measure the thermal sensitivities of both HCF samples. We employ a narrow linewidth laser (1558 nm, RIO Orion from Nufern), which is frequency locked to a carrier-envelope offset (CEO) stabilized optical frequency comb to eliminate the laser carrier frequency drift, which ensures that the measured phase response is only due to the thermally induced phase change in the fibers inside the interferometer. The fibers under test (thinly coated HCF, 24 m; dual-coated HCF, 30.2 m) were coiled with a diameter of 14 cm and spliced into a Mach-Zehnder interferometer made of a 2×2 input coupler and a 3×3 output coupler, both made of SSMFs with tails shorter than 1.5 m. These tails were cut to the same length and placed together to minimize any relative change in an optical phase between them. The 3×3 output coupler enables unambiguous phase change extraction, including its sign [12], which is not possible when a 2×2 coupler is used. We did not use any polarization control in the interferometer, since a drift in polarization that causes change of the interference contrast was accounted for in data processing. We put the interferometer into a thermal chamber and placed a thermistor in the center of the coiled fiber under test.

Figure 2(a) shows the measured accumulated phase change in response to temperature change. For the thinly coated HCF, the phase change follows the temperature change (e.g., when the temperature became stable, the accumulated phase was constant), and the extracted phase sensitivity (calculated from the phase difference at the three temperature points) is shown in Fig. 3. The situation is different for the dual-coated HCF, Fig. 2(b), where the phase keeps changing, even when the temperature is stable. Following a temperature rise (20 $\rightarrow$ 38°C and then 38 $\rightarrow$ 59°C), the measured phase initially rose, but then continued to reduce for several hours. When decreasing the temperature, we observe the same effect but with an opposite sign. The drift of the phase at constant temperature eventually stopped when the temperature was kept constant. In Fig. 2(b), we see this for example at the temperature of 59°C (time of 12–22 h) and at 20°C (time of 36–60 h). It looks as if the coating was relaxing over time, reducing its effect on the overall fiber length change. We will discuss this effect in detail later. Besides this time-dependent behavior, the overall thermal sensitivity of the dual-coated HCF is larger than that for the thinly coated HCF, as shown in Fig. 3, in which the phase measured at five

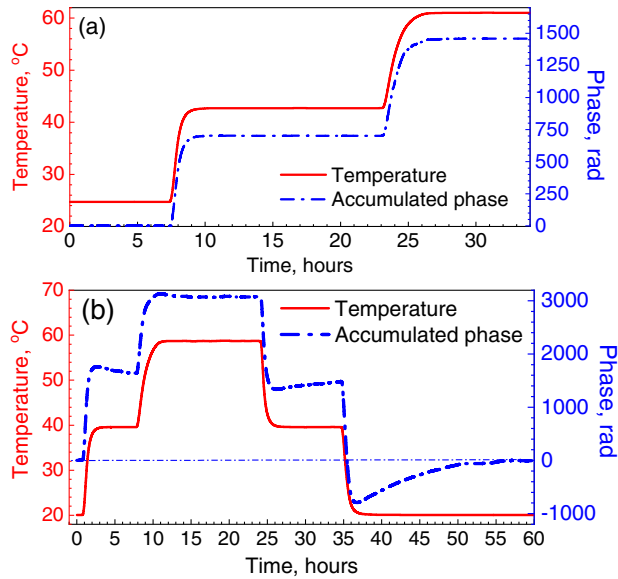


Fig. 2. Temperature inside the thermal chamber and accumulated phase measured for the (a) 24 m long thinly coated HCF and (b) 30 m long dual-coated HCF.

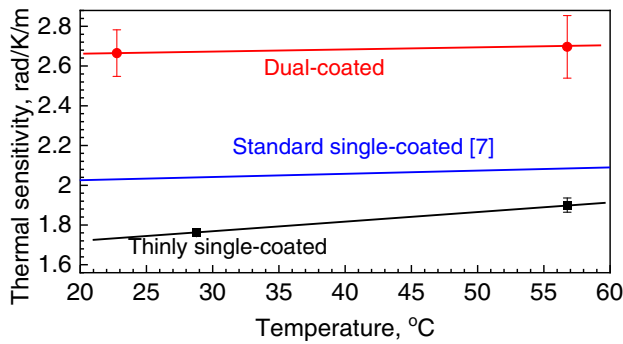


Fig. 3. Phase thermal sensitivity of three measured HCFs (thinly coated, standard single-coated and dual-coated) calculated from data shown in Fig. 2 (thinly coated and dual-coated) and from the data in [6] (standard single-coated).

stable temperature points in Fig. 2(b) is used for the calculation. The error bars correspond to the difference between the values measured when increasing and decreasing the temperature. For comparison, we also plot in Fig. 3 the data adopted from [6] obtained for our typical single-coated HCF.

We see that the thermal sensitivities of the two single-coated HCFs differ by about 20%, with the thinly coated HCF achieving a lower sensitivity result than expected from the uncoated HCF (2.2 rad/m/K, as discussed earlier). Inserting the thinly coated HCF thermal sensitivity and the above-mentioned coating parameters and fiber dimensions into Eq. (4) gives the silica coefficient of thermal expansion to be 0.3 ppm/K (at 25°C). This value is significantly lower than the widely used value of 0.55 ppm/K [5]. Even when neglecting the coating influence in Eq. (4), the value is 0.39 ppm/K, which is still significantly lower than expected. This is not in contradiction with values reported in the literature, where fused silica glass has been demonstrated to have a thermal expansion coefficient at room temperature spanning a relatively large range from 0.3 to

0.55 ppm/K [13–15]. This result is also consistent with our previous measurements [6] shown in Fig. 3, in which we measured a value of 0.47 ppm/K (already below that of 0.55 ppm/K), even for a single-coated HCF with standard coating thickness. This result is also consistent with our measurement on a dual-coated HCF, where using 0.3 ppm/K in Eq. (4) gives $\alpha_{\text{dual-coated-fib}} = 0.68$ ppm/K, which corresponds to the measured value (0.66 ppm/K, calculated from data in Fig. 3). We conclude that the silica coefficient of thermal expansion for optical fibers (in particular, HCF) can be as low as 0.30 ppm/K.

In Fig. 2(b), we show that, besides a higher value of thermal phase sensitivity, the HCF coating is also responsible for the phase relaxation effect, where the light phase keeps changing many hours after the temperature has been stabilized. To better understand this effect, we modeled the thermal behavior of the dual-coated HCF sample in COMSOL Multiphysics. The diameters of the coating and silica cladding of the modeled fiber were set to be the same as the manufactured sample, while the contribution from the inner tubes (microstructure) to the thermal sensitivity was neglected. In COMSOL, two independent physics interfaces (solid mechanics and heat transfer in solids) were employed and coupled together to simulate the thermal elongation of an optical fiber, including both heat transfer and the mechanical properties. In particular, we modeled the coating layers as viscoelastic materials, which are characterized (besides the elastic properties) by a relaxation time that describes the relaxation of thermal stress [16]. In the simulations, we considered the temperature observed in the experiment [shown in Fig. 2(b)]. We fitted all the parameters from their initial values obtained in the literature and relevant datasheets, in particular, Young's modulus, thermal expansion coefficients, and relaxation time. Figure 4 compares the measured [shown in Fig. 2(b)] and fitted phase changes in response to temperature change. We were able to obtain an excellent fit, especially in terms of the relaxation caused by the coating viscoelastic properties. It is worth mentioning that we were not able to obtain a good fit with $\alpha_{\text{silica}} = 0.55$ ppm/K, especially in terms of the relaxation. However, using $\alpha_{\text{silica}} = 0.3$ ppm/K as discussed above, allowed us to achieve a good fit, as shown in Fig. 4. The fitted time constant of the coating varies between 60 000 s at 25°C and 6000 s at 60°C. We consider this reasonable, as the other coating parameters (Young's modulus, in particular) also change significantly with temperature. To avoid this thermal relaxation induced phase shift, we suggest using a thick HCF silica cladding jacket in conjunction with a thin coating. Such a design also reduces the thermal phase sensitivity of the HCF,

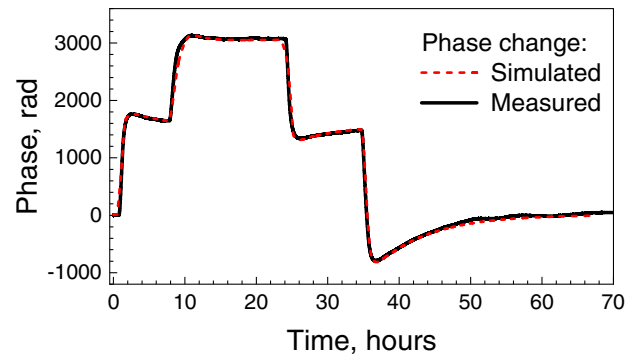


Fig. 4. Comparison of measured (shown in Fig. 2) and simulated thermal sensitivity of a dual-coated HCF considering viscoelastic behavior of the coating.

as we show here experimentally and as was previously demonstrated numerically [8]. A thick HCF silica jacket should also have the additional benefit of reducing the acoustic sensitivity of the HCF [17]. Another alternative to reduce the thermal relaxation effect is to use different coating materials that do not show as strong viscoelastic behavior such as acrylates, e.g., possibly polyimide.

In conclusions, we have shown experimentally that coatings can play a significant role in determining the thermal sensitivity of a HCF. We believe this is behind most of the discrepancies in the literature in terms of the thermal sensitivity measured for various HCFs. To minimize its unwanted effect, we fabricated a relatively thick (185 μm diameter) HCF with only 10 μm thin acrylate coating. The thin coating did not have any adverse effect on the mechanical strength of the fiber, which was kept coiled at 14 cm diameter over three months. Its thermal sensitivity was lower than expected and we believe this is because the thermal expansion coefficient of silica glass from which the HCF is made is lower than the value often used in the literature. This makes the HCF even less thermally sensitive (1.25 times) than the best values previously reported at room temperature [3], making it almost 30 times less sensitive than SSF (48 rad/m/K, as calculated from the data in [18]).

Finally, we showed that, besides compromising the phase sensitivity coefficient, the coating can also cause a drift in the optical phase, even when kept at constant temperature for a period, and we show that this is due to the coating's viscoelastic properties. Besides HCFs, this effect is expected to play a role also in SSF-based systems, e.g., in delayed-line interferometers for laser stabilization [19].

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Disclosures. The authors declare no conflicts of interest.

Data Availability. The data for this Letter are accessible through the University of Southampton Institutional Research Repository [20].

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